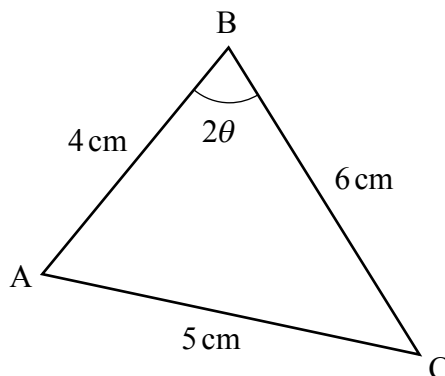


4. [Maximum mark: 6]

The following diagram shows triangle ABC , where $AB = 4\text{ cm}$, $BC = 6\text{ cm}$, $AC = 5\text{ cm}$ and $\hat{ABC} = 2\theta$.

diagram not to scale



Find the exact value of $\cos \theta$, giving your answer in the form $\frac{p\sqrt{2}}{q}$, where $p, q \in \mathbb{Z}^+$.

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